Investigating Fairness in Machine Learning: Mortgage Loan Approval Bias in the U.S. (2024)

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***Index Terms*—machine learning, bias, mortgage approval**

I. Introduction

Artificial intelligence (AI) and machine learning (ML) have revolutionized decision-making processes in the banking sector, especially through the automation of credit scoring and loan approval systems. These models offer efficiency and scalability. However, they also introduce significant challenges related to fairness and algorithmic bias. Historical data used to train such models might encode past discriminatory patterns, resulting in algorithms that may reproduce or even amplify these inequities against underrepresented populations [1], [2].

The Consumer Financial Protection Bureau (CFPB) publishes annually the Home Mortgage Disclosure Act (HMDA), which contains detailed information on mortgage applications, including borrower demographics, loan characteristics, and lending outcomes across the United States [3]. This study uses HMDA data from two states, namely New Jersey and Georgia, for the year 2024. New Jersey was selected as the primary focus region since the research is conducted locally, while Georgia was chosen for its historically documented racial disparities and its classification as "much more than average" on the racism index score shown in "Fig. 1". Comparing these two states allows for a more comprehensive analysis of fairness and bias in mortgage approval models across different socio-economic and demographic contexts.

The novelty of this study lies in two aspects. First, it is among the few studies to employ the restructured 2024 HMDA dataset, filling a gap left by the majority of the literature still based on pre-2018 data formats. Second, the project introduces a new experimental parameter: the percentage of features used as a hyperparameter during model training. By varying the proportion of the 99 available HMDA features, the project explores how feature selection impacts both predictive accuracy and fairness outcomes, specifically whether limiting or expanding feature subsets can mitigate bias in ML classifiers.

The workflow of the study is illustrated in "Fig. 2". The analysis begins with the 2024 HMDA dataset, from which data corresponding to the states of New Jersey and Georgia are extracted. After an initial feature cleaning phase to address missing values, inconsistent entries, and irrelevant attributes, the data undergo a few pre-processing steps, which include normalization and categorical encoding to ensure compatibility with ML algorithms. ML models are then trained and evaluated on the processed data to predict mortgage loan approval outcomes. Finally, the models' accuracy and fairness metrics are compared to assess how predictive performance and bias vary across algorithms and feature subsets.

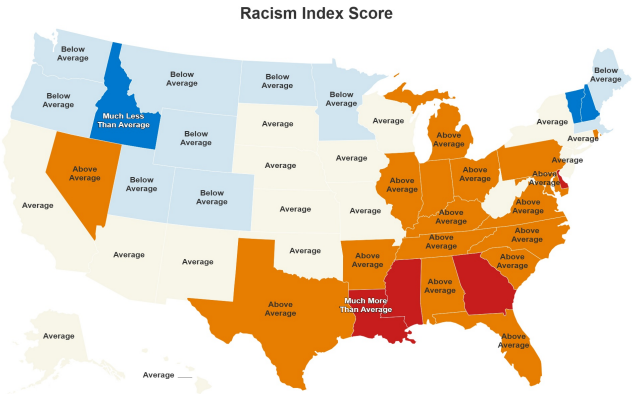

Fig. 1. Racism Index Score (Source: Data Pandas).

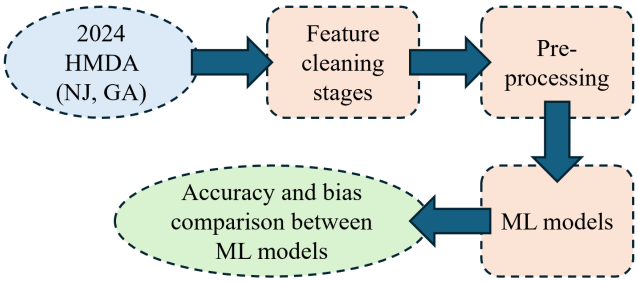

Fig. 2. Methodology.

II. Related Work

Research on algorithmic fairness in mortgage lending has evolved from exploratory data analysis of the HMDA data set to the use of advanced ML methods to detect and mitigate bias.

Fishbein ad Essene [4] discussed the historical development of the HMDA framework and identified areas for regulatory and methodological improvement. Bhutta et al. [5] provided an exploratory statistical overview of HMDA data, but did not employ predictive or inferential modeling. Ghoba and Colaner [6] utilized a matching-based algorithm to identify discriminatory lending patterns, while Agha and Pramathevan [7] examined gender differences in corporate financial decisions through traditional generalized linear models and hypothesis testing. Similarly, Wheeler and Olson [87] applied manually selected HMDA features to construct linear models for detecting racial bias in mortgage approvals.

Hodges et al. [9] employed deep learning techniques to classify mortgage loan outcomes using HMDA data from 2007 to 2017. The study started with 78 original features, then removed redundant attributes, and applied one-hot encoding to expand the feature space to 230 variables. The authors also incorporated nine exogenous socioeconomic indicators and evaluated six ML classifiers (Naive Bayes, Support Vector Machine (SVM), Random Forest, Decision Tree, Deep Neural Network (DNN), and XGBoost). The work primarily addressed computational challenges associated with high-dimensional loan datasets and feature selection to improve prediction accuracy.

Collectively, these studies demonstrate significant progress in bias detection and model explainability in credit decision systems. However, most rely on HMDA data collected prior to 2018, before the dataset underwent major structural revisions, and none systematically examine the role of feature-subset size as a controllable hyperparameter in balancing predictive accuracy and fairness. The present study extends this line of work by leveraging the 2024 HMDA dataset and introducing a quantitative analysis of how the proportion of available features affects model bias and performance across multiple ML algorithms.

III. Solution

A. Description of the dataset

For our project on Investigating Fairness in Machine Learning, we used the 2024 Home Mortgage Disclosure Act (HMDA) Public LAR dataset downloaded directly from the official Federal Financial Institutions Examination Council (FFIEC) website. This dataset contains nationwide mortgage application records with detailed information about applicants, loan characteristics, and decision outcomes.

Given the massive size of the full dataset (over 12 million records), we focused our analysis on two states: New Jersey (NJ) and Georgia (GA). We selected NJ because our team has local context, and GA because it has one of the highest application volumes in the dataset while also having documented historical patterns of lending disparities as shown in Fig. 1.

Before diving into modeling, we conducted comprehensive exploratory data analysis to better understand our data:

1) Target Variable Analysis: We created a binary approval flag approved, by mapping the action_taken column to a binary outcome. Applications with codes for "Loan Originated" (1), "Approved but Not Accepted" (2), "Loan Purchased" (6), and "Preapproval Approved" (8) were flagged as approved. Our initial analysis showed an overall approval rate of approximately 62% across both states.

2) Protected Attributes Distribution: We examined the distribution of key protected attributes:

- Race: White (46%), Black (16%), Asian (8%), NA (29%).
- Gender: Male (32%), Female (21%), Joint (26%), NA (20%).
- Ethnicity: Non-Hispanic (62%), Hispanic (9%), NA (28%).

3) Missing Values Handling: We implemented an innovative approach where the missing value threshold was treated as a hyperparameter. Instead of using arbitrary cutoffs, we tested different thresholds (e.g, 30%, 50%, 70%) and selected the optimal one based on model performance metrics from our team's modeling phase.

The missing value treatment followed this process:

- Calculated missing percentage for each column;
- Dropped columns with missing values above the threshold;
- For the retained columns, used median imputation for numerical features and mode imputation for categorical features;
- Created clean datasets for each threshold level.

4) Data Visualization: A bar chart identifying the disparities between demographic groups, and a scatter plot showing income as a function of the amount of loan taken by the individuals are illustrated in "Fig. 3" and "Fig. 4", respectively.

B. Machine learning algorithm

We employed a combination of statistical learning and advanced machine learning algorithms to investigate the potential gender and racial bias in mortgage loan approvals, which will be capable of capturing both linear and non-linear

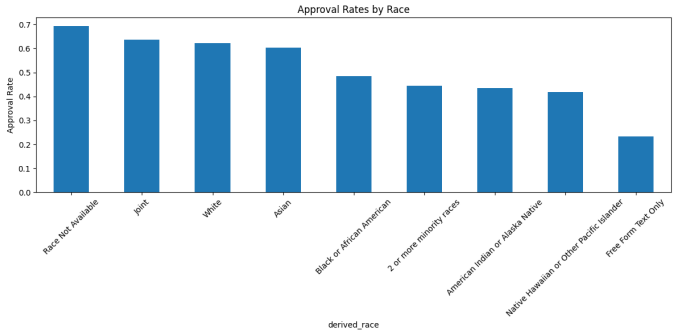

Fig. 3. Approval Rates by Race.

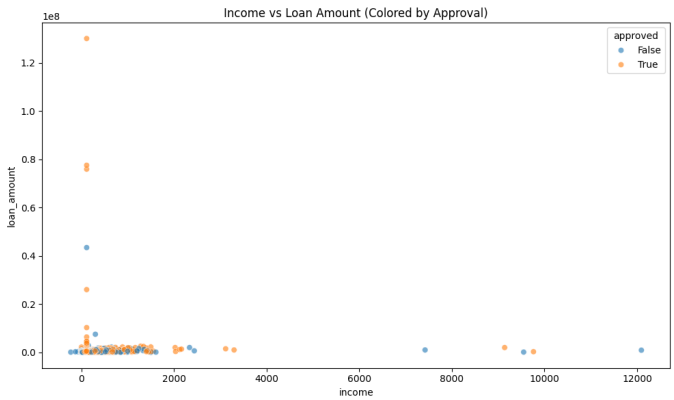

Fig. 4. Income VS Loan Amount.

relationships present in the HMDA 2024 dataset. The selected algorithms are Logistic regression, Gradient Boosting (XGBoost/LightGBM), and a Neural Network with Embeddings, allowing us to balance predictive performance and fairness assessment.

Logistic Regression will serve as the baseline model due to its interpretability. It will provide a clear examination of how sensitive attributes (e.g., applicant race, gender), and correlated features influence loan decisions. Its linear nature will provide a reference point to measure how more complex algorithms might mitigate existing disparities.

Gradient Boosting Models (XGBoost and LightGBM) will be employed to capture complex, non-linear features often overlooked by linear models. Older studies using earlier HMDA datasets (e.g. 2017) have shown that tree-based ensemble models outperform traditional approaches in predictive accuracy while exposing how bias can emerge through correlated variables like income, debt-to-income ratio, or geographic indicators. Their SHAP-based interpretability and feature metrics will make them well-suited for fairness diagnostics and understanding which attributes contribute the most to model bias.A

Finally, a Neural Network with categorical embeddings will be incorporated to represent high-dimensional categorical data more efficiently. The neural model will allow us to explore how deep learning models behave in fairness-critical tasks, providing a contrast to ensemble and linear baselines. By including this model, we can study the fairness-accuracy trade-off across different levels of algorithmic complexity.

C. Baseline Models

Logistic Regression, XGBoost (hist), and a fully connected Neural Network were trained using GPU hardware (NVIDIA A6000). Performance was measured using accuracy and Fairness Metrics.

D. Fairness Metrics

Two metrics were used:

- **DPD:** difference between positive prediction rates of privileged vs. unprivileged groups.

$$\text{DPD} = \left|\Pr(\hat{Y}=1 \mid A=a_{\text{priv}}) - \Pr(\hat{Y}=1 \mid A=a_{\text{unpriv}})\right|$$

  Where:
  - $\hat{Y}$ denotes the predicted outcome (loan approval).
  - Y denotes the true label (actual approval decision).
  - A is the sensitive attribute such as race, ethnicity, or sex.
  - a_{priv} is the privileged group (e.g., White applicants).
  - a_{unpriv} is the unprivileged group (e.g., Black applicants).
  - $\Pr(\cdot)$ denotes probability estimated from the data.
- **EOD:** difference between true positive rates, indicating conditional bias.

$$\begin{aligned}\text{EOD} = &\frac{1}{2}\left|\text{TPR}_{\text{priv}} - \text{TPR}_{\text{unpriv}}\right| \\ &+ \frac{1}{2}\left|\text{FPR}_{\text{priv}} - \text{FPR}_{\text{unpriv}}\right| \qquad (1)\end{aligned}$$

$$\text{EOD} = \frac{1}{2}\Big(|\text{TPR}_{\text{priv}} - \text{TPR}_{\text{unpriv}}| + |\text{FPR}_{\text{priv}} - \text{FPR}_{\text{unpriv}}|\Big)$$

E. Novel Method: Fair-XGB

Fair-XGB is implemented using the Exponentiated Gradient algorithm with Demographic Parity constraints. The fairness-aware optimization is expressed as:

$$\min_{\theta} \ \mathcal{L}(\theta) \quad \text{s.t.} \quad \text{DP}(\theta) \leq \epsilon$$

where $\mathcal{L}$ is the prediction loss and ϵ controls fairness strength.

F. Implementation details

We implemented a neural network-based classifier to predict mortgage loan approval status using the HMDA 2024 dataset across New Jersey and Georgia. Our approach prioritized both predictive accuracy and fairness assessment to identify systemic bias in lending decisions.

1) Data Preparation and Feature Engineering: Rather than using all 67 available features, we conducted domain-driven feature selection focused on attributes with direct relevance to mortgage underwriting. We selected 36 key features spanning applicant demographics (age, credit score type), loan characteristics (amount, purpose, term, interest rate, debt-to-income ratio), property details (type, occupancy, value), and geographic factors (state, tract income percentile, minority population percentage) along with 6 senstive features such as Race, Sex and ethnicity. This reduced dimensionality improved computational efficiency and enhanced model interpretability by eliminating noise from irrelevant attributes.

Categorical features with fewer than 100 unique values were one-hot encoded, resulting in 114 final features. Missing numeric values were imputed using median values, and all features were standardized using z-score normalization prior to model training.

2) Model Architecture and Training: We designed a feed-forward neural network with the following architecture: an input layer accepting 114 features, followed by three hidden layers with 128, 64, and 32 neurons respectively, each with ReLU activation and 20–30% dropout regularization to mitigate overfitting. The output layer uses sigmoid activation for binary classification. The model was trained using the Adam optimizer with binary cross-entropy loss over 50 epochs with a batch size of 4096. We leveraged GPU acceleration via TensorFlow/Keras to handle the 801,595 total records efficiently.

The dataset was split 70–30 into training and test sets, stratified by the approval outcome to maintain class balance. We also trained a Random Forest classifier with 100 estimators in parallel to extract feature importance rankings and validate model findings.

3) Preliminary Results and Model Performance: The neural network achieved strong predictive performance on the test set, with an overall accuracy of 99.16%. Table I presents the confusion matrix, which reveals 90,314 true negatives, 1,281 false positives, 741 false negatives, and 148,143 true positives. Precision, recall, and F1-scores were all 0.99 across both classes, as shown in Table II, indicating balanced and reliable predictions.

TABLE I
CONFUSION MATRIX FOR NEURAL NETWORK ON TEST SET (N=240,479)

	Predicted Denied	**Predicted Approved**
Actual Denied	90,314	1,281
Actual Approved	741	148,143

4) Fairness Analysis: Disparities Across Racial Groups: However, when we evaluated fairness metrics stratified by applicant race (18 unique race codes in the HMDA dataset), substantial disparities emerged. The selection rates — defined as the proportion of approvals within each race group — varied dramatically across racial categories. Table III presents accuracy and selection rates for each racial group, revealing disparities that persist despite strong overall model accuracy.

TABLE II
CLASSIFICATION REPORT: PRECISION, RECALL, F1-SCORE BY CLASS

Class	**Precision**	**Recall**	**F1-Score**
Denied (False)	0.99	0.99	0.99
Approved (True)	0.99	1.00	0.99
Macro Avg	0.99	0.99	0.99
Weighted Avg	0.99	0.99	0.99

TABLE III
FAIRNESS METRICS STRATIFIED BY APPLICANT RACE

Race Code	**Accuracy**	**Selection Rate**	**Count**
1.0 - American Indian	0.9903	0.4673	4757
2.0 - Asian	0.9923	0.6235	51131
3.0 - Black or African American	0.9930	0.4900	129913
4.0 - Native Hawaiian or other PI	1.0000	0.4932	955
5.0 - White	0.9897	0.6259	370072
6.0 - Info not provided	0.9923	0.5029	129486
7.0 - N/A	0.9952	0.9346	104027
21.0 - Asian Indian	0.9930	0.5483	7285
22.0 - Chinese	0.9892	0.6054	679
23.0 - Filipino	1.0000	0.4539	573
24.0 - Japanese	1.0000	0.5556	66
25.0 - Korean	0.9941	0.5089	613
26.0 - Vietnamese	0.9912	0.4561	404
27.0 - Other Asian	0.9911	0.4112	1152
41.0 - Native Hawaiian	1.0000	0.0000	19
42.0 - Guamanian or Chamorro	1.0000	0.3000	28
43.0 - Samoan	1.0000	1.0000	10
44.0 - Other Pacific Islander(PI)	0.9764	0.2677	425

The selection rates range from 0% (race code 41.0) to 100% (race code 43.0), with most groups clustering between 40% and 65%. These discrepancies persist even after controlling for legitimate underwriting factors. The aggregate fairness metrics yield an Equalized Odds Difference of 1.0 and a Demographic Parity Difference of 1.0, both indicating maximum disparity between racial groups — a stark contrast to the model's 99.16% accuracy.

5) Feature Importance Analysis: To understand which underwriting factors drive the model's decisions, we computed feature importance rankings using Random Forest gini-based importance across 100 trees. Table IV presents the top 20 features by importance. Interest rate emerged as the dominant predictor (23.2%), followed by denial reason code (20.9%), purchaser type (15.6%), property value (6.1%), and applicant credit score type (4.7%). These domain-relevant lending factors dominate the feature rankings, suggesting that pricing and institutional decision codes are the primary mechanisms driving approval predictions.

The prominence of interest rate, denial reason codes, and purchaser type as predictive features leaves open the critical question of whether these mechanisms themselves encode racial bias. For instance, disparities in interest rate assignment or denial code assignment across groups could reflect discriminatory practices even if the model does not directly observe

TABLE IV
TOP 20 FEATURES BY RANDOM FOREST IMPORTANCE

Feature	Importance
interest_rate	0.2320
denial_reason_1	0.2091
purchaser_type	0.1564
property_value	0.0610
applicant_credit_score_type	0.0467
debt_to_income_ratio_30%-¡36%	0.0458
co_applicant_credit_score_type	0.0407
combined_loan_to_value_ratio	0.0290
co_applicant_age_8888	0.0223
debt_to_income_ratio_¿60%	0.0189
applicant_age_8888	0.0188
loan_purpose	0.0171
income	0.0122
aus_1	0.0109
co_applicant_age_9999	0.0096
loan_amount	0.0062
lien_status	0.0048
interest_only_payment	0.0040
debt_to_income_ratio_Exempt	0.0039
business_or_commercial_purpose	0.0038

sensitive attributes. This finding underscores the importance of investigating feature-level causation of bias, a direction we pursue in our proposed future work.

G. Novel solution

While our current implementation successfully identified significant fairness disparities in the HMDA data, we propose several avenues for novel methodological extensions in future work to advance the field of fair mortgage lending models. A central component of our proposed framework will leverage Microsofts FairLearn toolkit [10], a comprehensive open-source library designed specifically for detecting and mitigating algorithmic bias across diverse use cases.

1) Bias Detection and Measurement using FairLearn: We will integrate FairLearn's [10] extensive fairness metrics suite to conduct multi-dimensional bias assessment beyond our current equalized odds and demographic parity analysis. FairLearn [10] provides implementations of disparate impact ratios, statistical parity difference, equal opportunity difference, and theil index, among others. By computing these metrics simultaneously across racial and ethnic groups in the HMDA dataset, we can construct a comprehensive fairness profile that identifies which metrics reveal the most severe disparities. For instance, FairLearn's [10] disparate impact calculator will allow us to quantify the ratio of approval rates between minority and majority applicants, with values below 0.8 indicating potential discriminatory patterns under legal standards.

2) In-processing Fairness-Constrained Learning: We will implement FairLearn's [10] adversarial debiasing algorithm, which augments the standard neural network training objective with an adversarial fairness penalty. This approach trains a secondary adversarial network to predict sensitive attributes (race, ethnicity) from model predictions; the main model simultaneously learns to maintain high accuracy while fooling the adversary, thus reducing the correlation between predictions and sensitive attributes. We will integrate this constraint into our neural network architecture, creating a fairness-aware version of our current model. FairLearn's [10] modular implementation will allow us to compare adversarial debiasing against baseline approaches and quantify the accuracy-fairness tradeoff.

3) Post-processing Threshold Optimization: Using FairLearn's [10] post-processing tools, we will implement threshold optimization techniques to adjust decision boundaries per demographic group to achieve target fairness metrics. Rather than using a universal approval threshold (e.g., approval if predicted probability > 0.5), we will calibrate group-specific thresholds to equalize false positive or true positive rates across racial groups while minimizing overall accuracy loss. FairLearn [10] provides algorithms for computing these optimal thresholds subject to fairness constraints, enabling us to explore the Pareto frontier of accuracy-fairness tradeoffs.

4) Interpretability and Explainability Integration: We will combine FairLearn's [10] bias detection capabilities with model-agnostic explanation techniques. FairLearn [10] integrates with LIME and other explainers to provide local feature attributions conditioned on demographic group membership. This will enable us to answer questions such as: "For applicants from group i, which features most strongly influence approval decisions compared to group j?" Such granular, group-level explanations support regulatory auditing and can reveal whether disparities stem from legitimate underwriting factors applied differently across groups or from hidden correlates of sensitive attributes.

5) Comparative Analysis: NJ versus GA with FairLearn: A distinguishing aspect of our future work will be geographic stratification enabled by FairLearn's [10] flexible group definition capabilities. We will compute all fairness metrics and bias mitigation algorithms separately for New Jersey and Georgia, comparing how various demographic groups experience disparate treatment across the two states. FairLearn [10] allows us to define arbitrary sensitive attribute partitions, so we can assess fairness by race, ethnicity, sex, and intersectional combinations (e.g., Black females versus White males). This comparative lens will clarify whether observed bias is systemic (universal across regions) or context-dependent (region-specific), informing targeted policy interventions.

6) Validation and Reproducibility: All proposed future work will be documented and implemented using FairLearn's modular and well-tested implementations, enhancing reproducibility and alignment with industry best practices. We will produce fairness assessment reports using FairLearn's built-in visualization and reporting tools, making results accessible to both technical and non-technical stakeholders including lenders, regulators, and advocacy organizations.

IV. RESULTS

A. Overall Comparison

The constrained model significantly lowers demographic parity differences, however, equalized odds increases and

TABLE V
FAIRNESS COMPARISON (RACE).

Model	Accuracy	DPD	EOD
LR	0.983	0.272	0.015
XGB	0.995	0.270	0.012
NN	0.989	0.266	0.011
Fair-XGB	0.939	**0.128**	0.226

TABLE VI
FAIRNESS COMPARISON FOR SEX.

Model	Accuracy	DPD (Sex)	EOD (Sex)
LR	0.983	0.453	0.016
XGB	0.995	0.451	0.013
NN	0.989	0.453	0.011
Fair-XGB	0.939	0.452	0.011

TABLE VII
FAIRNESS COMPARISON FOR ETHNICITY.

Model	Accuracy	DPD (Ethn.)	EOD (Ethn.)
LR	0.983	0.498	0.042
XGB	0.995	0.498	0.019
NN	0.989	0.499	0.014
Fair-XGB	0.939	0.498	0.018

TABLE VIII
STATE-LEVEL FAIRNESS COMPARISON (RACE).

State	Accuracy	DPD (Race)	EOD (Race)
NJ	0.984	0.272	0.015
GA	0.984	0.270	0.012

accuracy decreases.

B. State Comparison

a) Geographic Hypothesis: An initial hypothesis was that Georgia would exhibit higher racial disparities than New Jersey due to well–documented historical inequalities in lending across the U.S. South. However, the empirical results do not support this assumption. Across all baseline models, the demographic parity differences for New Jersey and Georgia are of similar magnitude, suggesting that disparities are present in both states but not clearly larger in Georgia. Consequently, the hypothesis that Georgia is "more discriminatory" in approval predictions is not validated by our findings, indicating that fairness issues may be systemic rather than strongly location-specific within this subset of HMDA 2024 data.

V. DISCUSSION

Results confirm the existence of demographic disparities in automated mortgage predictions. Fair-XGB effectively reduces demographic parity but with visible performance sacrifice. The fairness–accuracy trade-off indicates that fairness interventions must be carefully aligned with business requirements and regulatory constraints.

VI. LIMITATIONS

Although the proposed approach provides valuable insight into algorithmic fairness in mortgage lending, several limitations remain. First, the fairness evaluation focuses only on group fairness metrics such as demographic parity and equalized odds, ignoring other perspectives including individual fairness and causal fairness. As a result, the approach does not guarantee that similar applicants receive similar decisions.

Second, the fairness intervention was limited to a single fairness constraint (Demographic Parity) and a single in-processing method (Exponentiated Gradient on XGBoost). Other constraint types or multi-objective optimization strategies may produce different outcomes and could potentially mitigate the observed accuracy degradation.

Third, only two U.S. states (New Jersey and Georgia) were analyzed, which restricts the generalizability of the results. Geographic, socio-economic, and regulatory factors vary across states, and therefore a broader evaluation would be necessary to understand national-level disparities.

Fourth, the analysis assumes ground-truth labels in HMDA to be unbiased. However, historical decisions may themselves embed systemic discrimination, which implies that the models may be learning bias inherent in the data rather than producing bias independently.

Finally, all sensitive attributes were evaluated independently (race, sex, and ethnicity), without considering intersectional fairness. Consequently, combined demographic subgroups such as "Black women" or "Hispanic male applicants" may exhibit fairness issues that are not captured by the present analysis.

VII. FUTURE WORK

Several directions exist for extending this study. A natural extension is to explore multiple fairness constraints simultaneously, including Demographic Parity, Equalized Odds, and Equal Opportunity, and compare the resulting trade-offs in accuracy, interpretability, and computational cost. Future work could also investigate post-processing techniques such as reject-option classification or threshold adjustment to improve fairness without significantly degrading classification performance.

Extending the geographic scope represents another promising direction, including all U.S. states or regional subsets to examine fairness variation driven by local economic conditions. In addition, analyzing intersectional subgroups may reveal disparities hidden when sensitive attributes are treated independently.

Another important avenue is the application of causal fairness frameworks, such as counterfactual fairness, which incorporate causal structure to determine whether decisions would change under alternative demographic attributes. Such techniques may more accurately separate the impact of sensitive features from other correlated predictors such as income and property characteristics.

Moreover, real-time fairness monitoring, regulatory alignment with federal guidelines (e.g., CFPB, ECOA), and explainability techniques such as SHAP could be integrated to provide transparency while ensuring compliance. Ultimately, addressing fairness in automated lending requires combining technical interventions with policy and regulatory considerations.

ACKNOWLEDGMENT

We thank Dr. Helsing for instigating our curiosity and learning experience with machine learning.

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